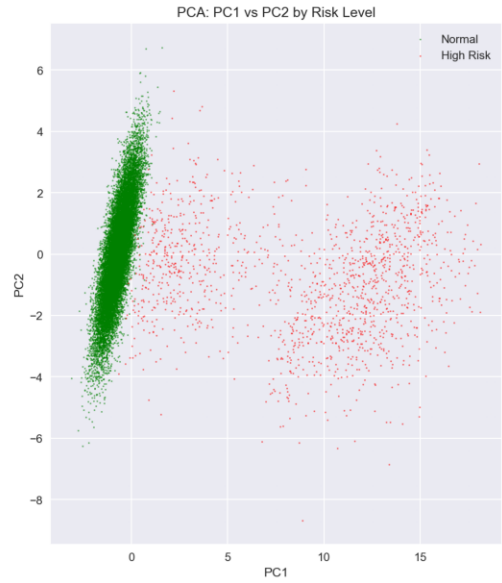
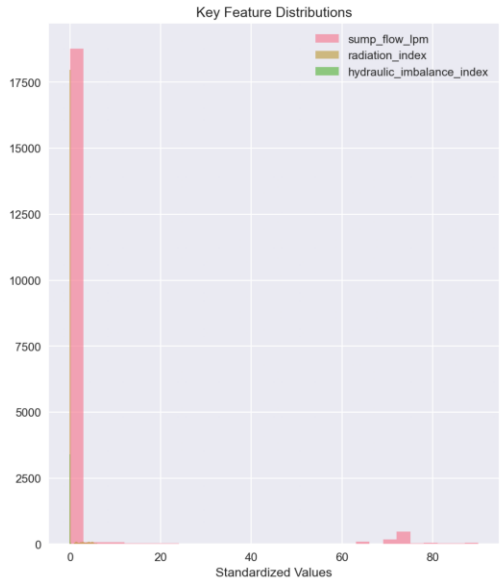
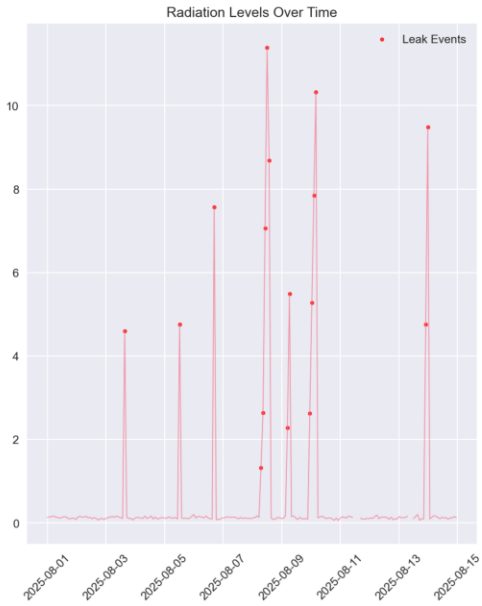
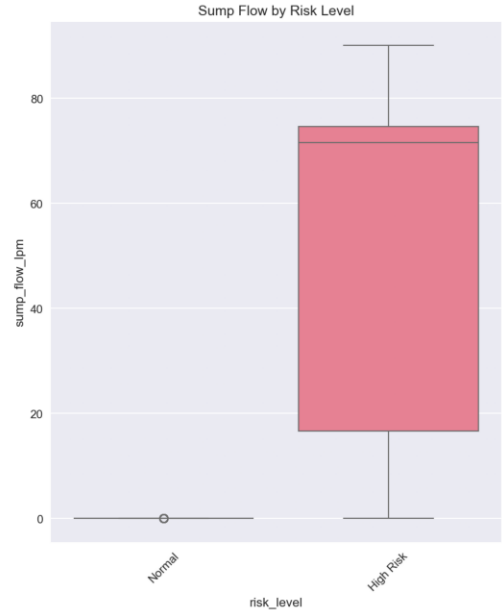
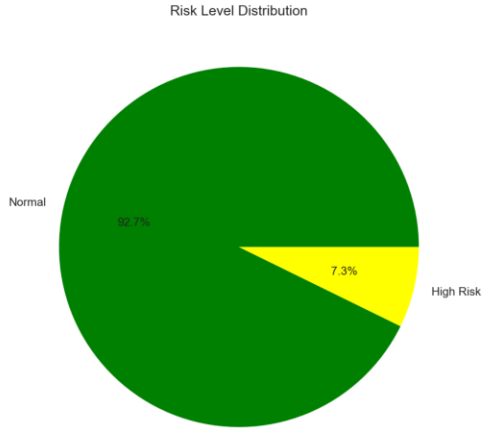
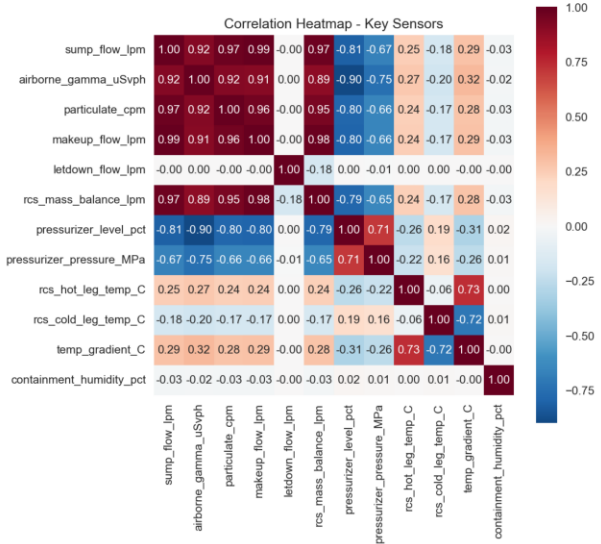


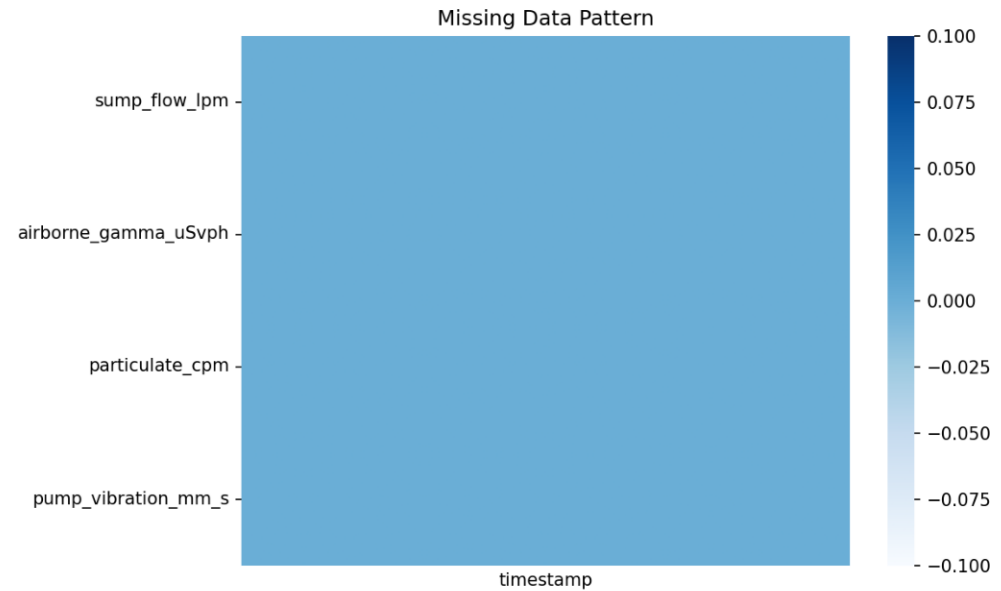
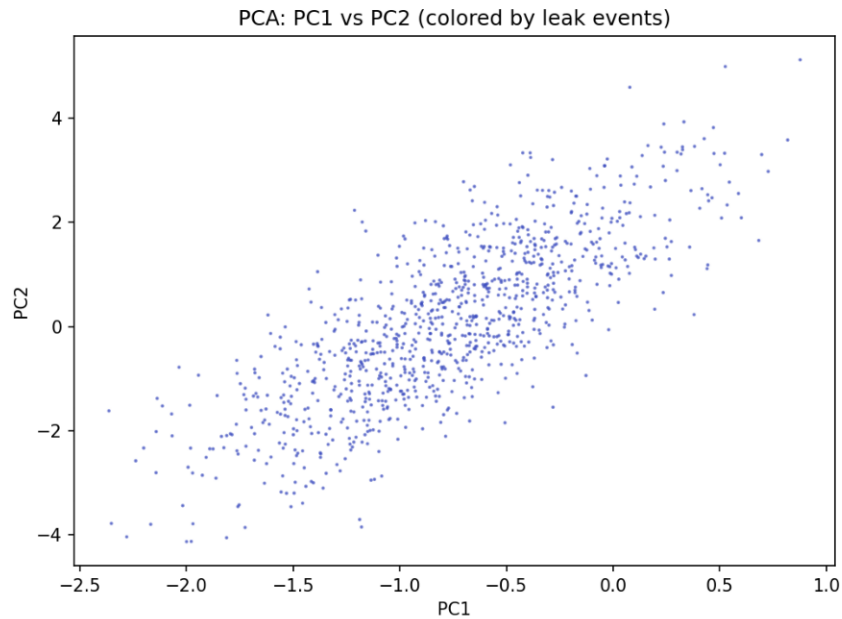
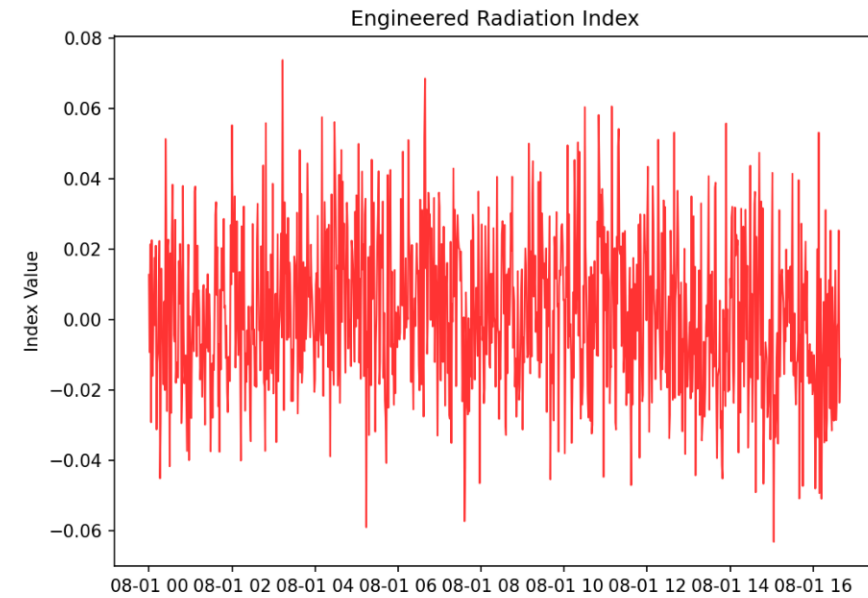
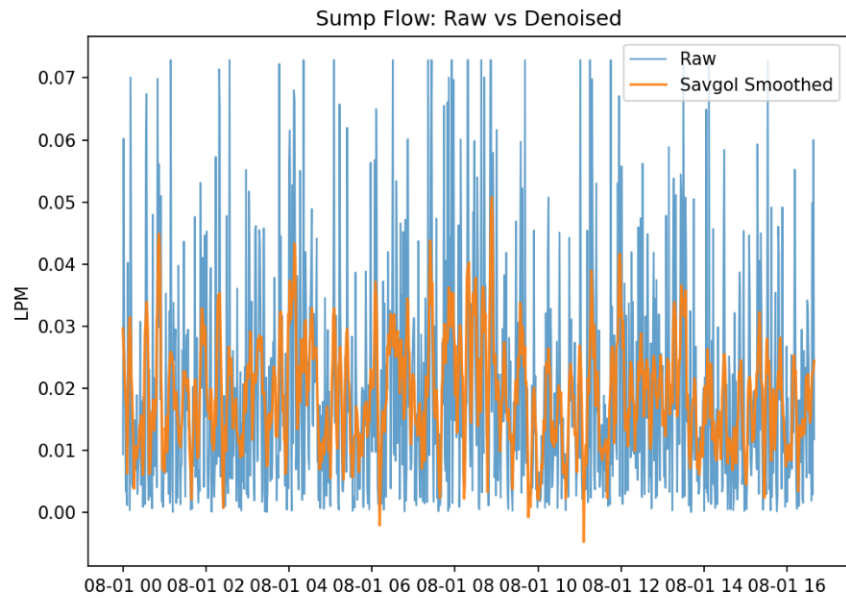
Nuclear Plant Leak Detection — Results Summary

This PDF collects the key result figures generated by the pipeline without moving any files. Each page embeds an existing image from the data/mining_results or data/processed directory, with a short caption summarizing the insight.

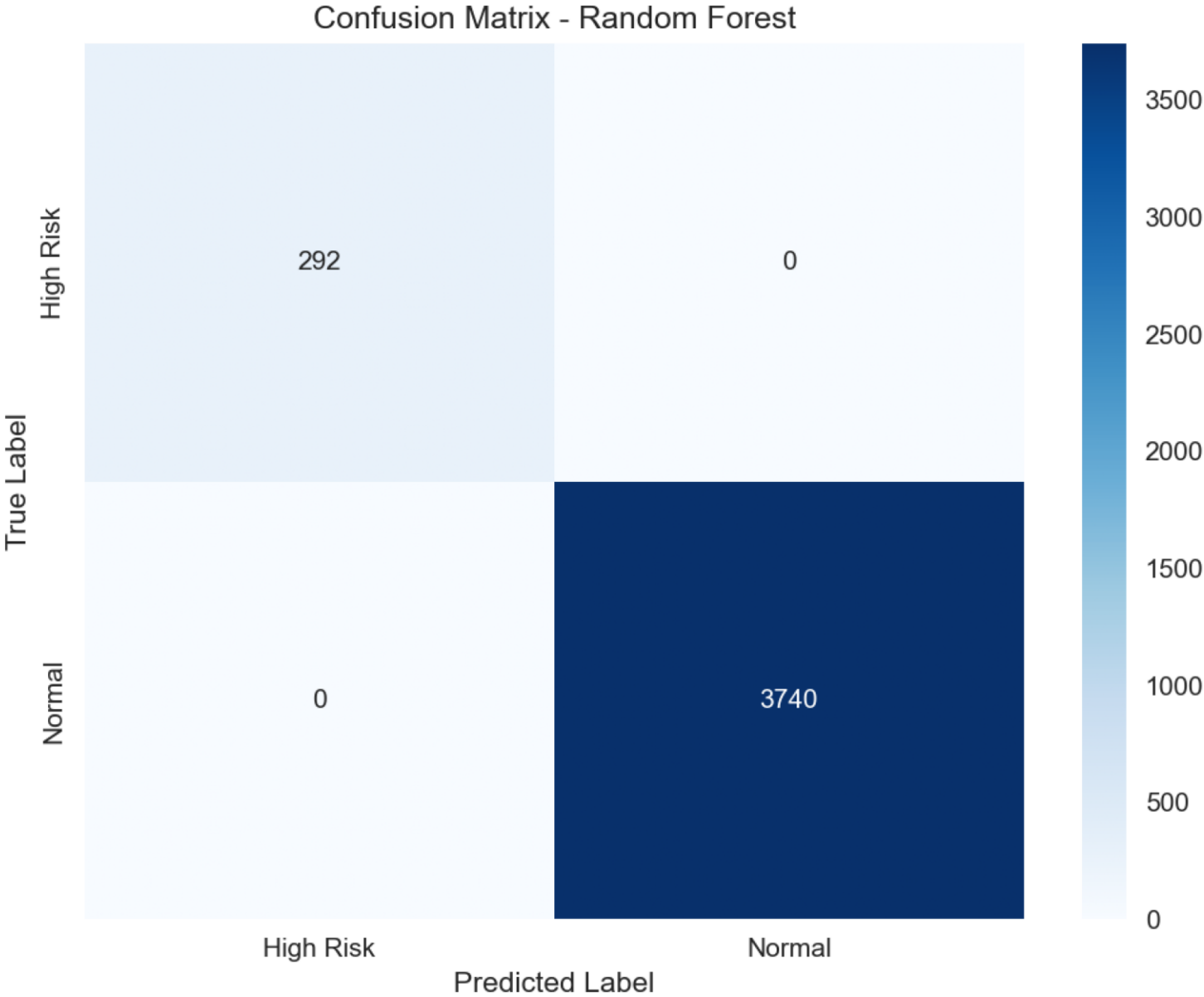
Exploratory Data Analysis



Processing Overview

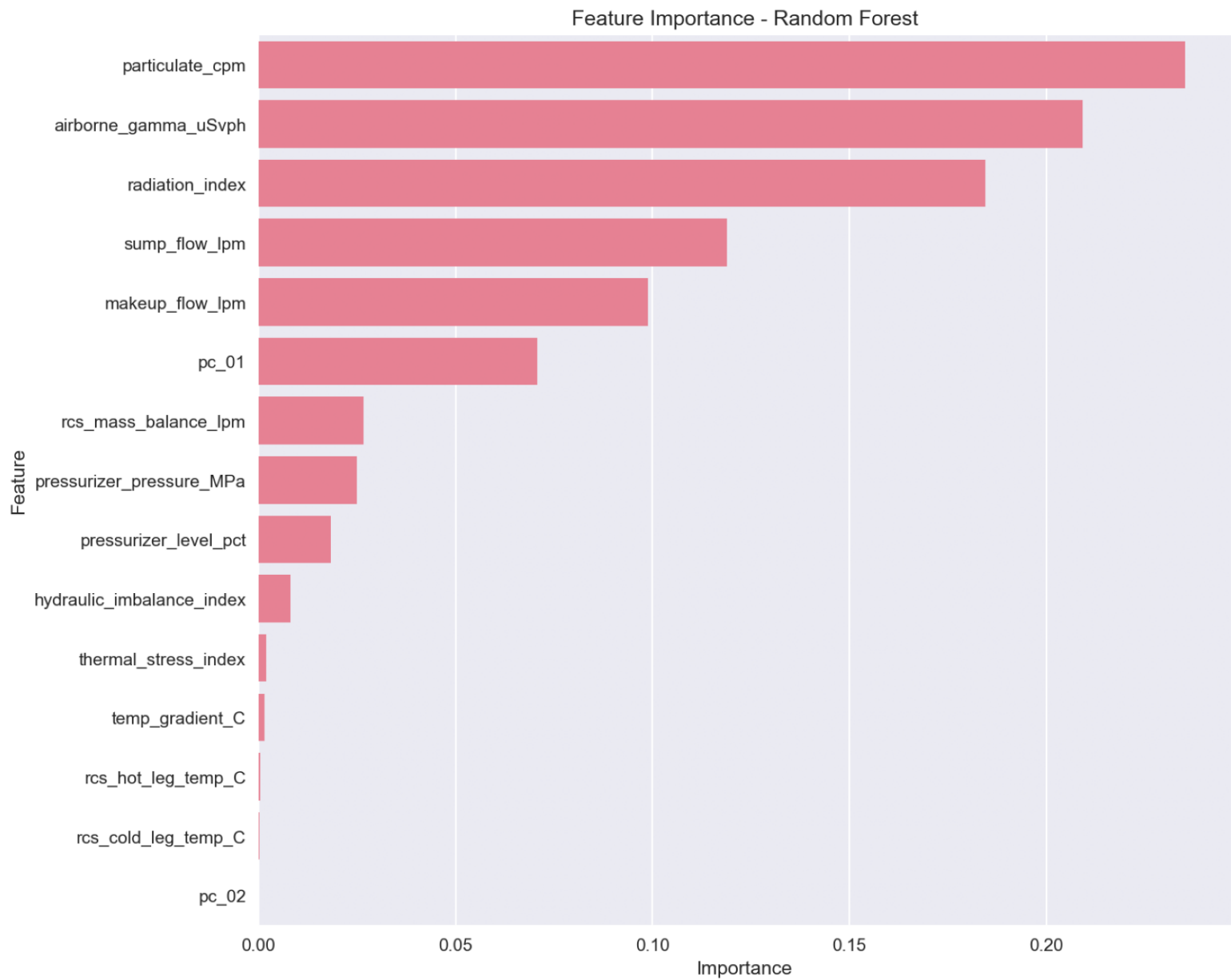


Multi-class Classification: Confusion Matrix (Random Forest)

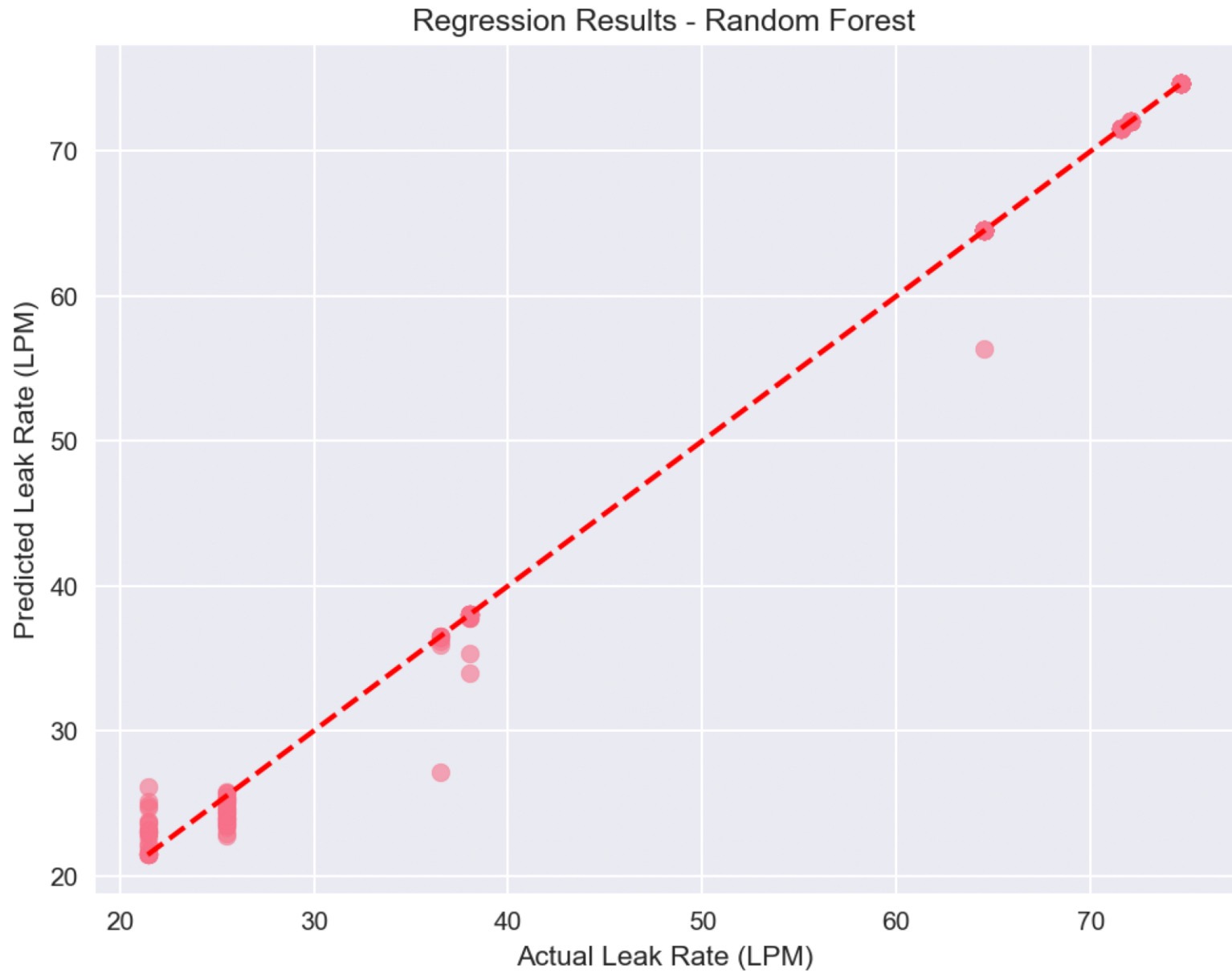


Perfect diagonal confirms near-perfect risk-level prediction on synthetic data.
File: data\mining_results\confusion_matrix_random_forest.png

Feature Importance (Random Forest)

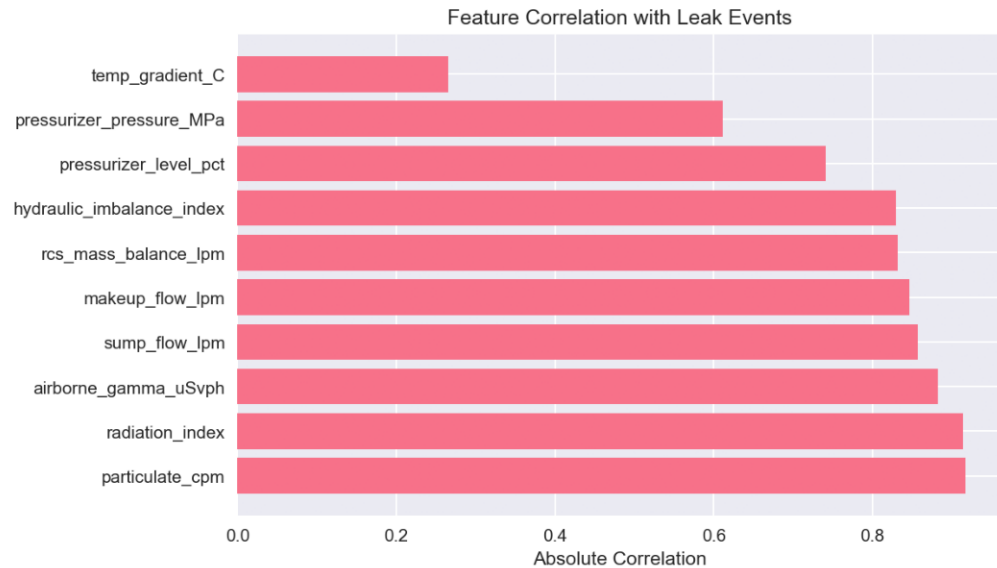
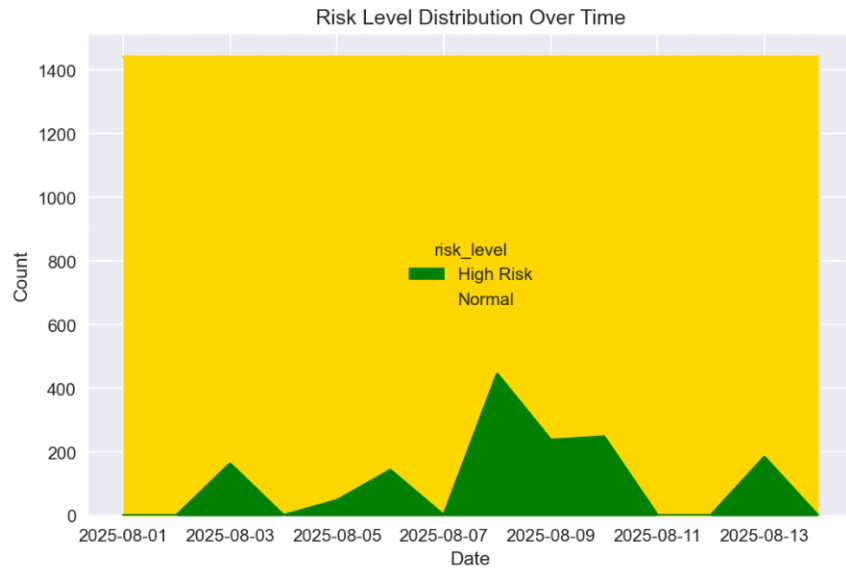
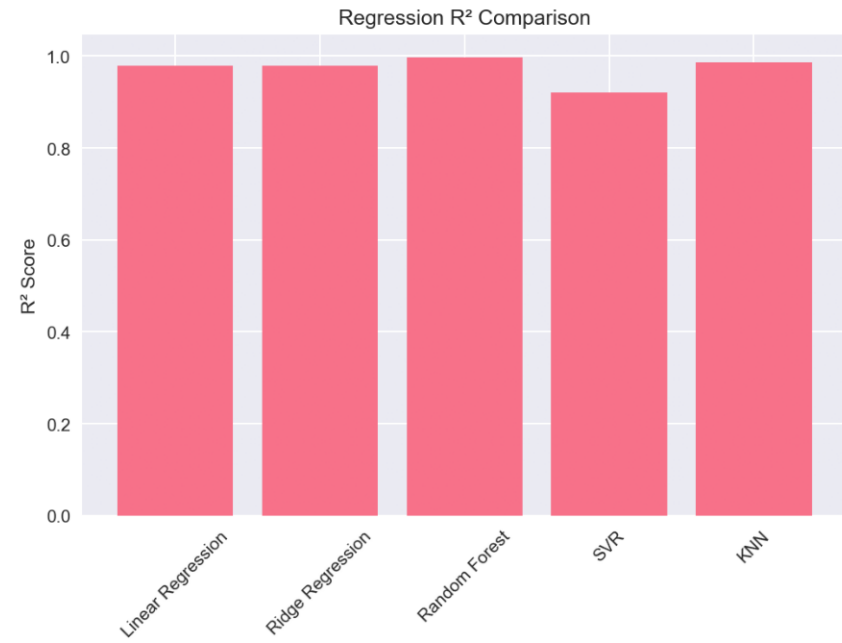
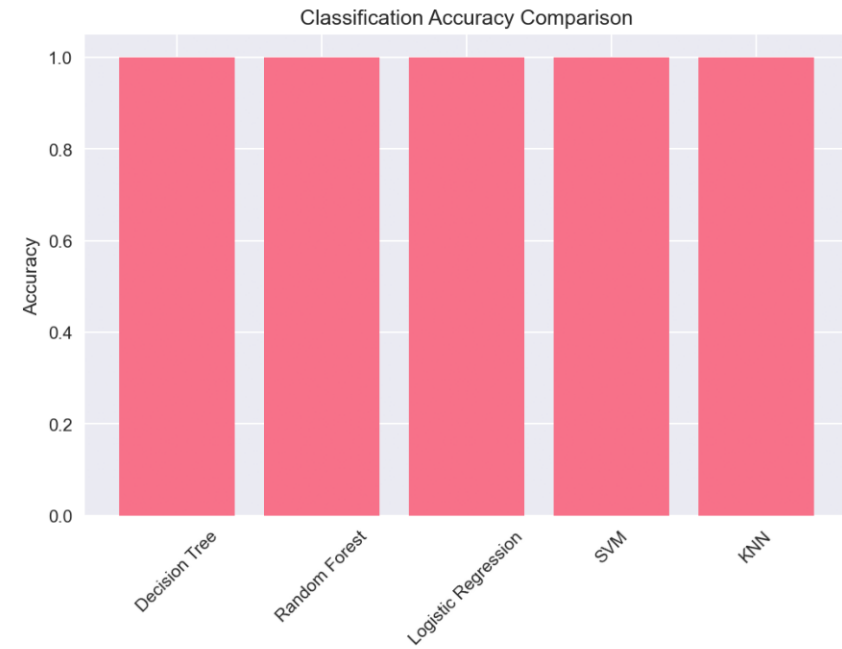


Regression: Actual vs Predicted Leak Rate (Random Forest)



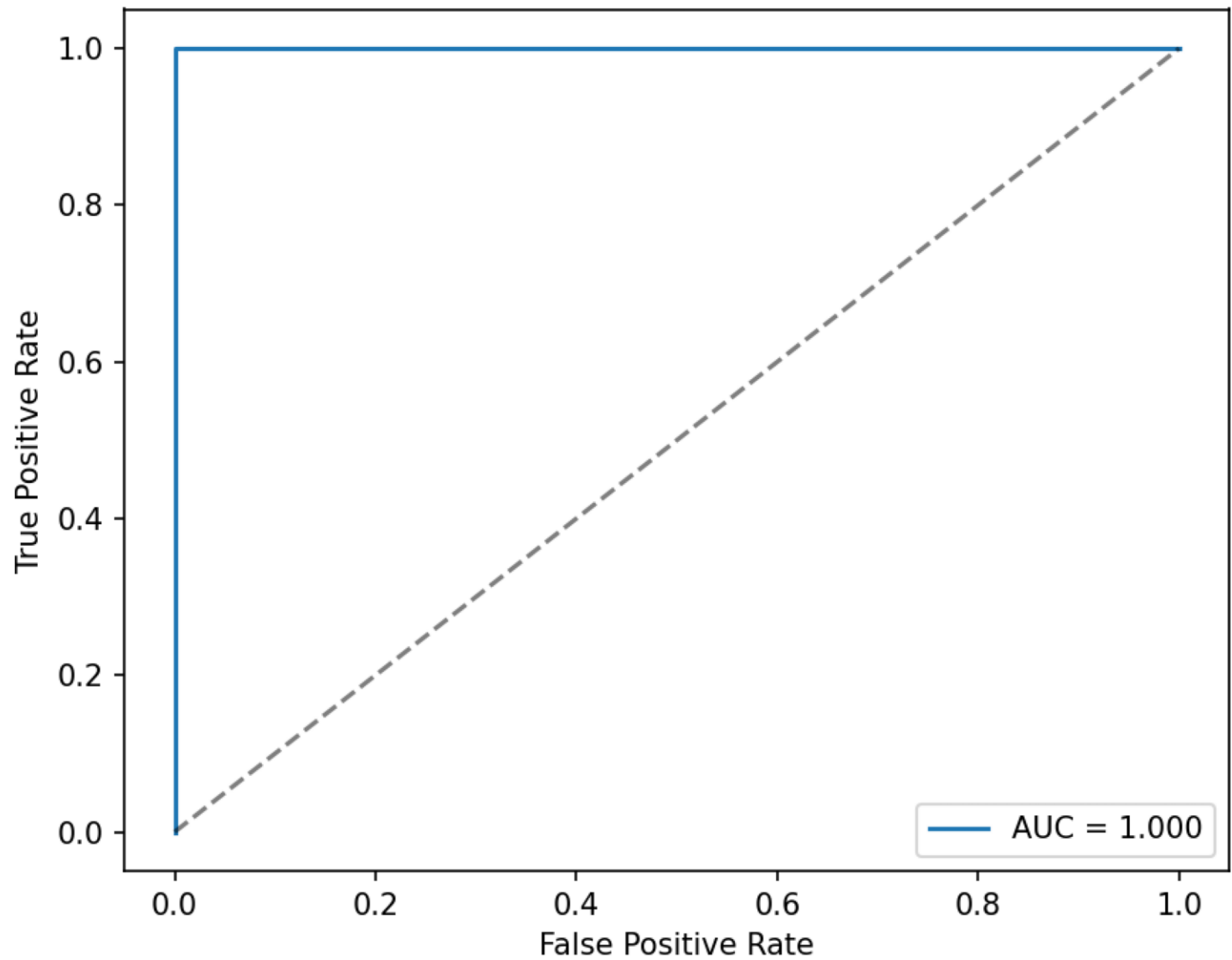
Tight alignment with the identity line; $R^2 \approx 0.997$ and $RMSE \approx 1.06$ LPM.
File: data\mining_results\regression_scatter.png

Model Comparison Summary



Binary Classification ROC (Random Forest)

ROC - Random Forest (Binary Leak Detection)



Dimensionality Reduction (PCA)

